

Attitude Change in Immersive Virtual Environments

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ABSTRACT

Immersive Virtual Environments (IVEs) are foreseen to function as effective tools of persuasion fundamentally because of the phenomenon of presence, the perception of and response to the virtual environment as real. Previous research suggests that IVE exposure decreases biases and fosters positive attitudes more effectively than traditional media even in the context of polarizing topics, for example migration and climate change. The research goals of the dissertation are to examine the influence of IVE experiences on attitudes and the factors moderating this effect. It does so in four consecutive research stages, building on a theoretical framework of media richness, presence, attitude extremity and motivated reasoning. Altogether, it a. provides an overview of extant empirical literature in a review with meta-analysis, b. explores the production, dissemination and evaluation of IVEs used by charitable organizations in a series of interviews and c. tests for the moderating role of attitude extremity in two experimental research designs.

Keywords: Virtual Reality, Presence, Attitudes, Attitude Extremity, Motivated Reasoning

1 INTRODUCTION

One key question about immersive virtual environments (IVEs) is if, and why, they influence the ways we think and behave in the physical world more effectively than traditional media. Central to this strand of research are attitudes, the evaluation of an object on a spectrum from negative to positive [1] and attitude change, the degree to which an attitude can be modified. Attitudes are vital for the democratic process, because they help citizens make choices about fundamental political practices, from political participation and voting decisions [2]. As media play a key role in the influence of attitudes [3], the advent of VR technology has led to an increasing number of research exploring the effects of IVEs. Fundamentally, it is the experience of a social problem in VR that makes it more tangible and concrete for users. Underlying this experience is the central construct of *presence* and its subtypes, namely the subjective experience of being there [4]. Building on this premise, practitioners recently have shown large interest in IVE investment. For example, in only four years time (2015 - 2019) the United Nations invested in 21 VR awareness campaigns in order to facilitate “positive social change” [5]. Over the past twenty years, the growing number of sociopolitically-themed VR projects has led to the general assumption that IVE experiences are effective for social campaigning purposes [6]. While academics and practitioners alike explore the potential of IVE interventions to foster positive attitudes towards issues like climate change [7], the present dissertation position argues that those are topics infused by moral and partisan preferences. It has been consistently demonstrated that attitudes of ideological, political and religious origin tend to be strong, stable and resistant to change [8], and that the exposure to counterattitudinal information can even amplify extreme attitudes for reasons of social identity protection [9]. Thus I argue that attitude extremity can decrease or increase the effect of IVE interventions, depending on the initial position. In sum, my thesis is led by three

overarching research questions: a. What factors play a role in attitude change in IVEs?, b. What is the main effect of presence on attitudes and c. To what extent does attitude extremity moderate this effect? I attempt to answer these questions in a cumulative dissertation that comprises of four research papers.

2 THEORY

2.1 Media Richness & Presence

Recent considerations have connected media richness with the persuasive effect of IVE experiences [10]. Media richness theory places media on a continuum from lean to reach, depending on the amount of cues, or data, it can transmit [11]. Because VR technology is capable to mediate a plethora of cues, users eventually experience a feeling of being enveloped in the artificial environment, unaware of the physical world [4]. This feeling is referred to as presence. As consistently demonstrated in the field of consumer behavior research [12], media that are richer tend to evoke larger persuasive effects. In the same vein, because VR can induce feelings of presence, the term has been a central variable in models of influence and persuasion in IVEs [13].

2.2 Attitude Extremity & Motivated Reasoning

It is characteristic for people with extreme attitudes to minimize the relevance of counterattitudinal information, whilst strongly endorsing information that supports their initial position [14]. Hence prior attitude extremity can potentially decrease, or increase, the effect of IVE induced presence on users' attitudes towards strongly contested and potentially polarizing issues. I draw this assumption based on recent scholarship demonstrating that even less immersive media, for example social media, can increase polarization when users are exposed to opposite views [15].

3 CUMULATIVE DISSERTATION OUTLINE

3.1 Paper 1: Attitude Change in VR: A systematic review and meta-analysis (ongoing)

In Paper 1 I explore the question if IVE interventions are more effective in changing attitudes than other traditional media, and elaborate on the theoretical constructs that factor into this effectiveness. I then review current findings according to topic of social object and offer first evidence-based answers based on the results of a meta-analysis. Finally, I aggregate patterns in methodology and provide an overview of relevant research design and equipment.

3.2 Paper 2: The use of VR (Virtual Reality) for awareness raising and charitable giving. Results from expert interviews (ongoing)

Paper 2 examine the conception, production, dissemination and evaluation of Virtual Reality (VR) videos used by charitable organizations. By directly interviewing experts involved in the field, the project aims at obtaining insights about the connection of immersive technologies and attitude-consistent behavior, thus expanding the findings of Paper 1.

Two subsequent experimental studies focus on the treatment effect of presence and its subtypes on attitudes and the potential moderating role of attitude extremity. Because it is out of the scope of the present position paper, I elaborate on the details of Paper 3.

3.3 Paper 3

Paper 3 examines a moderation model in which the effect of social presence on pro-social behavior towards migrants is moderated by prior attitude extremity. It does so in a pre-test post-behavioral treatment design with two conditions and one control. A power analysis will help determine the sample size. All groups will receive an online pre-test three weeks prior the interventions in order to avoid pre-treatment effects. The pre-test will assess sociodemographics, interest in documentaries (buffer questions) as well as selected items of the Symbolic Racism Scale [16]. On site subjects will be randomly assigned to a. VR stimulus with social presence, b. VR stimulus without social presence and c. control group. Thus the independent variable is social presence (the psychological state in which the virtual actors are experienced as actual actors in a sensory or a non-sensory way [4]), the dependent variable is pro-social behavior (response to the post-stimulus behavior treatment) and the moderating variable is attitude extremity (measured with the absolute deviation from the scale midpoint as indicator [17]).

4 CONCLUSION

Overall, I seek to contribute to the extant literature on persuasion in IVEs by first exploring main effects and then focusing on the role of attitude extremity. To my knowledge, there is little research on how the effectiveness of IVEs is moderated by prior extreme attitudes. While this does come with a practical relevance as it explores IVEs as tools for sociopolitical campaigning in an increasingly polarized social landscape, there are several challenges to the experimental studies that I want to address. Firstly, the isolation of causal effects. Because IVE interventions are psychophysiological experiences novel to subjects it is difficult to isolate treatment effects. In comparison to reading an article or watching a video, there is little control over the virtual objects that grasp users' attention in VR and their influence on the dependent variable. Secondly, sampling and ethics. To control for attitude extremity, the sample must be sufficiently large and heterogeneous. However, by including subjects with extreme attitudes and exposing them to counter-attitudinal or attitude-conform information, the project risks to contribute to attitude polarization. I seek to address and discuss both challenges, the isolation of causation and ethical considerations, during the Consortium.

REFERENCES

- [1] Bohner, Gerd & Dickel, Nina (2010). Attitudes and Attitude Change. *Annual Review of Psychology*, 62(1), 391-417. <https://doi.org/10.1146/annurev.psych.121208.131609>
- [2] Rosenstone, Steven J. & Hansen, John Mark (1993). *Mobilization, Participation and Democracy*. New York: Macmillan
- [3] Bryant, J., & Oliver, M. B. (Eds.). (2009). *Media Effects*. New York: Routledge. <https://doi.org/10.4324/9780203877111>
- [4] Lee, Kwan Min (2004). Presence, Explicated. *Communication Theory*, 14 (1), 27-50. <https://doi.org/10.1111/j.1468-2885.2004.tb00302.x>
- [5] UNVR (2020). VR Films. Accessed 30.11.2020 <http://unvr.sdgactioncampaign.org/vr-films/#.X8UYCs1KjDc>
- [6] Borawska, A., Borawski, M., & Łatuszyńska, M. (2018). The Concept of Virtual Reality System to Study the Media Message Effectiveness of Social Campaigns. *Procedia Computer Science*, 126, 1616–1626. <https://doi.org/10.1016/j.procs.2018.08.135>
- [7] Markowitz, David M., Laha, Rob, Perone, Brian P., Pea, Roy D. & Bailenson, Jeremy N. (2018). Immersive Virtual Reality Field Trips Facilitate Learning About Climate Change. *Frontiers in Psychology*, 9, Article 2364. <https://doi.org/10.3389/fpsyg.2018.02364>
- [8] Krosnick, J. A., & Petty, R. E. (1995). Attitude strength: An overview. In Petty, R.E. & Krosnick J.A. (Eds.), *Ohio State University series on attitudes and persuasion*, Vol. 4. *Attitude strength: Antecedents and consequences*, 1–24. Lawrence Erlbaum Associates, Inc.
- [9] Lodge, M., & Taber, C. S. (2013). *The Rationalizing Voter*. New York: Cambridge University Press.
- [10] Fox, Jesse, Christy, Kathryn R. & Vang, Mao H. (2014). The Experience of Presence in Persuasive Virtual Environments. In (Eds.) Riva, Giuseppe, Waterworth, John & Murray, Dianne *Interacting with Presence*. 164-178. Warschau/Berlin: Walter De Gruyter GmbH.
- [11] Daft, R. L., & Lengel, R. H. (1986). Organizational Information Requirements, Media Richness and Structural Design. *Management Science*, 32(5), 554–571. <https://doi.org/10.1287/mnsc.32.5.554>
- [12] Saat, Rafeah Mat & Selamat, Mohamad Hisyam (2014). An Examination of Consumer's Attitude towards Corporate Social Responsibility (CSR) Web Communication Using Media Richness Theory. *Procedia, Social and Behavioral Sciences*, 155, 392-97. <https://doi.org/10.1016/j.sbspro.2014.10.311>
- [13] Blascovich, Jim, Loomis, Jack, Beall, Andrew C., Swinith, Kimberly R., Hoyt Crystal L. & Bailenson, Jeremy N. (2002). Immersive Virtual Environment Technology as a Methodological Tool for Social Psychology. *Psychological Inquiry*, 13:2, 103-124. https://doi.org/10.1207/S15327965PLI1302_01
- [14] Kunda, Z. (1990). The case for motivated reasoning. *Psychological Bulletin*, 108(3), 480–498. <https://doi.org/10.1037/0033-2909.108.3.480>
- [15] Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. B. F., Lee, Jaemin, Mann, Marcus, Merhout, Friedolin & Volfovsky, Alexander (2018). Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37), 9216–9221. <https://doi.org/10.31235/osf.io/4ygux>
- [16] Henry, P. J., & Sears, D. O. (2002). The Symbolic Racism 2000 Scale. *Political Psychology*, 23(2), 253–283. <https://doi.org/10.1111/0162-895x.00281>
- [17] Downing, J. W., Judd, C. M., & Brauer, M. (1992). Effects of repeated expressions on attitude extremity. *Journal of Personality and Social Psychology*, 63(1), 17–29. <https://doi.org/10.1037/0022-3514.63.1.17>